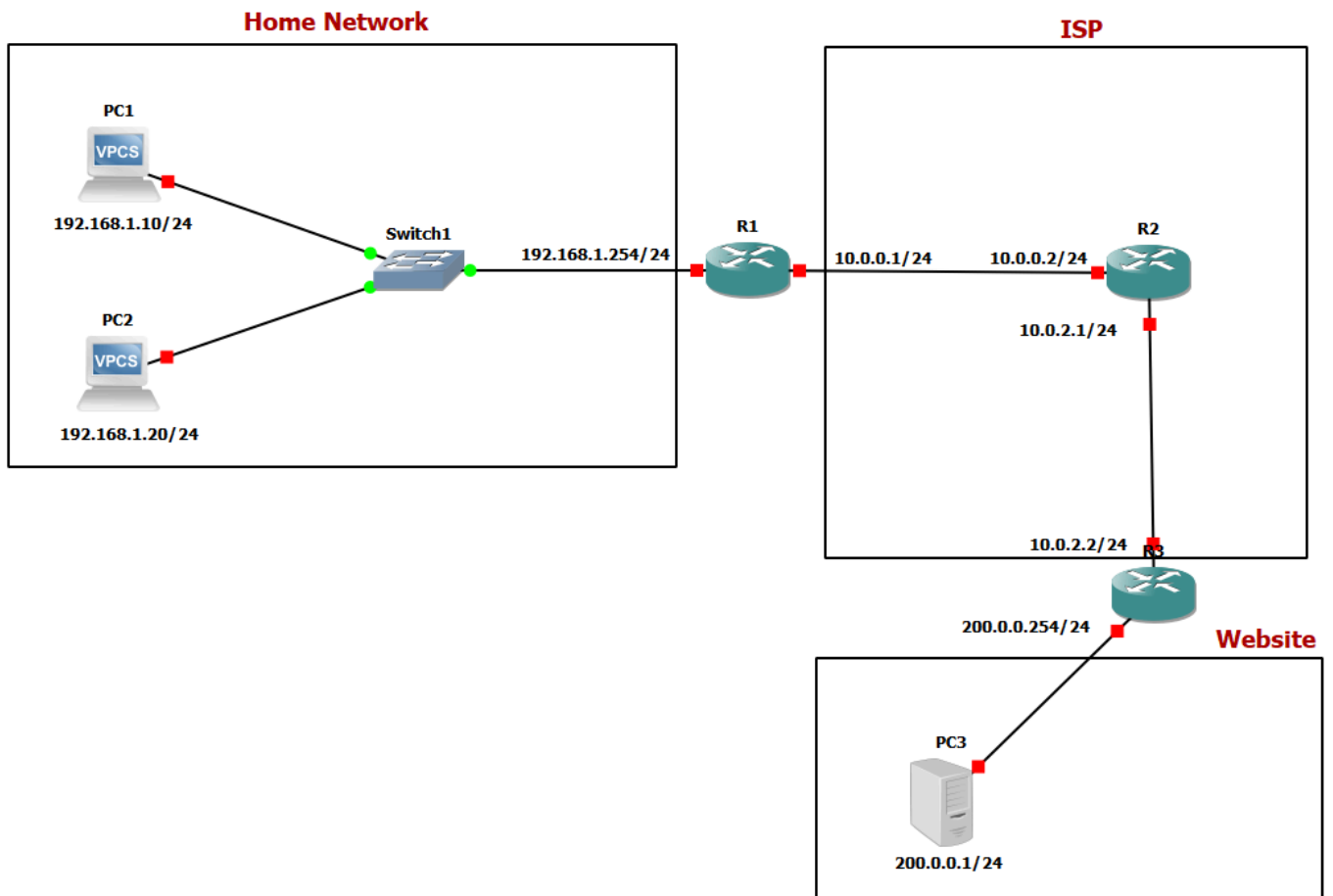


Part 1:-



Setting IP addresses to PC's and Routers

```
PC1> show ip  
  
NAME       : PC1[1]  
IP/MASK    : 192.168.1.10/24  
GATEWAY    : 192.168.1.254  
DNS        :  
MAC        : 00:50:79:66:68:00  
LPORT      : 10024  
RHOST:PORT : 127.0.0.1:10025  
MTU        : 1500
```

→ IP address and gateway set for PC1

```
PC2> show ip  
  
NAME       : PC2[1]  
IP/MASK    : 192.168.1.20/24  
GATEWAY    : 192.168.1.254  
DNS        :  
MAC        : 00:50:79:66:68:01  
LPORT      : 10026  
RHOST:PORT : 127.0.0.1:10027  
MTU        : 1500
```

→ IP address and gateway set for PC2

```
FastEthernet0/0 is up, line protocol is up  
  Hardware is Gt96k FE, address is c001.3040.0000 (bia c001.3040.0000)  
  Internet address is 192.168.1.254/24
```

```
FastEthernet0/1 is up, line protocol is up  
  Hardware is Gt96k FE, address is c001.3040.0001 (bia c001.3040.0001)  
  Internet address is 10.0.0.1/24
```

→ IP address set for Router 1

```
FastEthernet0/0 is up, line protocol is up  
  Hardware is Gt96k FE, address is c002.4950.0000 (bia c002.4950.0000)  
  Internet address is 10.0.0.2/24
```

```
FastEthernet0/1 is up, line protocol is up  
  Hardware is Gt96k FE, address is c002.4950.0001 (bia c002.4950.0001)  
  Internet address is 10.0.2.1/24
```

→ IP address set for Router 2

```
FastEthernet0/0 is up, line protocol is up
  Hardware is Gt96k FE, address is c003.4c1c.0000 (bia c003.4c1c.0000)
  Internet address is 10.0.2.2/24
```

```
FastEthernet0/1 is up, line protocol is up
  Hardware is Gt96k FE, address is c003.4c1c.0001 (bia c003.4c1c.0001)
  Internet address is 200.0.0.254/24
```

→ IP address set for Router 3

```
PC3> show ip

NAME       : PC3[1]
IP/MASK     : 200.0.0.1/24
GATEWAY     : 200.0.0.254
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 10028
RHOST:PORT  : 127.0.0.1:10029
MTU         : 1500
```

→ IP address and gateway set for PC3

→ Cannot ping Router2 because no routes were set on the routers, static or RIP.

Part 2:-

```
S    200.0.0.0/24 [1/0] via 10.0.0.2
    10.0.0.0/24 is subnetted, 2 subnets
S      10.0.2.0 [1/0] via 10.0.0.2
C      10.0.0.0 is directly connected, FastEthernet0/1
C      192.168.1.0/24 is directly connected, FastEthernet0/0
R1#
```

→ IP routes set for Router1 to network 200.0.0.0 and 10.0.2.0

```
S    200.0.0.0/24 [1/0] via 10.0.2.2
    10.0.0.0/24 is subnetted, 2 subnets
C      10.0.2.0 is directly connected, FastEthernet0/1
C      10.0.0.0 is directly connected, FastEthernet0/0
S      192.168.1.0/24 [1/0] via 10.0.0.1
R2#
```

→ IP routes set for Router2 to network 200.0.0.0 and 192.168.1.0

```

C    200.0.0.0/24 is directly connected, FastEthernet0/1
    10.0.0.0/24 is subnetted, 2 subnets
C      10.0.2.0 is directly connected, FastEthernet0/0
S      10.0.0.0 [1/0] via 10.0.2.1
S     192.168.1.0/24 [1/0] via 10.0.2.1
R3#

```

→ IP routes set for Router3 to network 10.0.0.0 and 192.168.1.0

```

PC2> ping 200.0.0.1
84 bytes from 200.0.0.1 icmp_seq=1 ttl=61 time=91.493 ms
84 bytes from 200.0.0.1 icmp_seq=2 ttl=61 time=90.688 ms
84 bytes from 200.0.0.1 icmp_seq=3 ttl=61 time=91.661 ms
84 bytes from 200.0.0.1 icmp_seq=4 ttl=61 time=91.529 ms
84 bytes from 200.0.0.1 icmp_seq=5 ttl=61 time=92.232 ms

```

→ Pinging to PC3 from PC1

→ Another way is to set Router1 and Router 3 IP routes to 0.0.0.0 0.0.0.0 10.0.0.2/
0.0.0.0 0.0.0.0 10.0.2.1 as all the communications from '**Home Network**' and '**Website**'
have only 1 router to hop which is Router2

Q1: if you only configure the default command above (and no other routes), would a ping from PC1 to R2 work?

→ No, You would have to supply IP routes to R2 to send the packet back to the network 192.168.1.0, and hence wouldn't work.

Part 3:-

```

R    200.0.0.0/24 [120/2] via 10.0.0.2, 00:00:19, FastEthernet0/1
    10.0.0.0/24 is subnetted, 2 subnets
R      10.0.2.0 [120/1] via 10.0.0.2, 00:00:19, FastEthernet0/1
C      10.0.0.0 is directly connected, FastEthernet0/1
C     192.168.1.0/24 is directly connected, FastEthernet0/0

```

→ R1 has the networks advertised by R3 and

```

R    200.0.0.0/24 [120/1] via 10.0.2.2, 00:00:04, FastEthernet0/1
    10.0.0.0/24 is subnetted, 2 subnets
C    10.0.2.0 is directly connected, FastEthernet0/1
C    10.0.0.0 is directly connected, FastEthernet0/0
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:22, FastEthernet0/0

```

→ R2 has the networks advertised by R1 and R3 and

```

C    200.0.0.0/24 is directly connected, FastEthernet0/1
    10.0.0.0/24 is subnetted, 2 subnets
C    10.0.2.0 is directly connected, FastEthernet0/0
R    10.0.0.0 [120/1] via 10.0.2.1, 00:00:00, FastEthernet0/0
R    192.168.1.0/24 [120/2] via 10.0.2.1, 00:00:01, FastEthernet0/0

```

→ R3 has the networks advertised by R1 and

```

PC1> ping 200.0.0.1
84 bytes from 200.0.0.1 icmp_seq=1 ttl=61 time=92.168 ms
84 bytes from 200.0.0.1 icmp_seq=2 ttl=61 time=92.529 ms
84 bytes from 200.0.0.1 icmp_seq=3 ttl=61 time=90.991 ms
84 bytes from 200.0.0.1 icmp_seq=4 ttl=61 time=91.367 ms
84 bytes from 200.0.0.1 icmp_seq=5 ttl=61 time=91.613 ms

```

→ Ping from PC1 to PC3 works!!

190	101.503069	10.0.0.2	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xc3d6, seq=16/4096, ttl=255 (request in 189)
191	102.559104	192.168.1.10	10.0.0.2	ICMP	98 Echo (ping) request	id=0xc4d6, seq=17/4352, ttl=63 (reply in 192)
192	102.574130	10.0.0.2	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xc4d6, seq=17/4352, ttl=255 (request in 191)
193	103.238787	10.0.0.2	224.0.0.9	RIPv2	86 Response	
194	103.629282	192.168.1.10	10.0.0.2	ICMP	98 Echo (ping) request	id=0xc5d6, seq=18/4608, ttl=63 (reply in 195)
195	103.645388	10.0.0.2	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xc5d6, seq=18/4608, ttl=255 (request in 194)
196	104.700877	192.168.1.10	10.0.0.2	ICMP	98 Echo (ping) request	id=0xc6d6, seq=19/4864, ttl=63 (reply in 197)
197	104.715880	10.0.0.2	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xc6d6, seq=19/4864, ttl=255 (request in 196)

→ We notice that there is a red packet with the protocol “RIPv2” within the capture this means that the router is advertising its routes to other routers so it knows where to route the packets to.